

Dinh Thai Tuan

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in [SisypheanHUS](#) |  [SisypheanHUS](#)

Hanoi, Vietnam

OBJECTIVE

Bachelor in Applied Mathematics with experience in alpha research, data analysis, machine learning, and academic research. Interested in quantitative research, machine learning, and factor investing, with a strong background in probability theory, mathematical modelling for data-driven applications.

EDUCATION

- **Vietnam National University – University of Science** 
Bachelor of Science in Applied Mathematics
◦ GPA: 7.3/10
◦ Main courses: Differential Equations, Calculus, PDEs, Probability Theory, Math Modelling, Statistics.
◦ Thesis: [Raman Spectroscopy for Methanol Detection](#) 
◦ Advisor: [Dr. Vu Duong](#)  – Institute of Physics, VAST
Aug. 2022 – Sep. 2026
Hanoi, Vietnam
- **Nguyen Duy Trinh High School**
Secondary Education
2019 – 2022
Nghị Loc, Nghe An, Vietnam

EXPERIENCE

- **VinSmart Future**
AI Engineer OJT
Apr. – Jul. 2026
Hanoi, Vietnam
◦ Studying and evaluating machine learning approaches for fraud detection, including XGBoost and Random Forest, using the PaySim dataset.
- **Vietnam Institute for Advanced Study in Mathematics** 
Summer Research Student
Aug. – Nov. 2025
Hanoi, Vietnam
◦ Research in epidemic modeling and actuarial science under the supervision of Dr. Hoang Manh Tuan.
◦ Contributor of a manuscript on a modified SIS model for health insurance pricing.
- **Group of Mathematical Biology, Nara Women's University** 
Summer Research Student
Aug. 2025
Nara, Japan
◦ Short summer research program on dynamical systems in population biology.
◦ Developed and presented a SIR modeling project in R.
- **Institute of Physics, Vietnam Academy of Science and Technology** 
Research Intern
July 2025 – Present
Hanoi, Vietnam
◦ Proposed a deep learning-based approach for quantifying methanol concentration in ethanol-methanol mixtures using Raman spectroscopy.
◦ Proposed and evaluated 1D and 2D CNN-based methods for the classification of methanol and ethanol from Raman spectral data.
◦ First author of three conference papers: one paper for ICAEP-9 (2025) and 2 paper for ICPA-NCOS 14 (2026, incl. one oral).
- **WorldQuant BRAIN** 
Research Consultant
Apr. 2024 – Present
Hanoi, Remote
◦ Submitted 1,800+ alphas across global equity markets using Fast Expression Language and Python; built automated pipelines for template mining, correlation-based selection, and Power Pool allocation.
◦ 5-time Grandmaster (top 2%) and 3-time Master (top 8%) in the Genius Program, out of 10,000+ consultants.
◦ Exclusive delegate from Vietnam – among 12 selected globally – for the IQC 2025 finals in Singapore.

PROJECTS

- **RamanSisyphean – Spectral Identification**
Tools: Python, JavaScript
2026

◦ Browser-based material identification tool via Raman, fluorescence, and absorption spectroscopy using Bayesian spectral matching (16,000+ reference spectra, 2,500 classes). Achieved 90.3% top-1 accuracy on LOO evaluation.

SKILLS

- **Programming Languages:** Python, R, Fast Expression Language, $\text{\LaTeX}$
- **Libraries & Frameworks:** TensorFlow, PyTorch, scikit-learn, NumPy, Pandas
- **Knowledge Areas:** Portfolio Construction, Backtesting, Factor Analysis, Machine Learning, Deep Learning, Feature Engineering, Model Validation

PUBLICATIONS

- **Dinh, T. T.** Duong, V. (2025). Raman spectroscopy for methanol detection: 1D vs. 2D CNN performance evaluation. *Proc. ICAEP-9*.
- **D. T. Tuan**, N. T. Hieu, A. I. Vodchits, V. A. Orlovich, P. H. Minh V. Duong (2026). Raman spectroscopic identification of chemicals using a closed-form Bayesian marginal likelihood. *ICPA-NCOS 14*, oral presentation.
- **D. T. Tuan**, N. T. Hieu, P. H. Minh V. Duong (2026). Raman peak matching for component detection in compounds. *Proc. ICPA-NCOS 14*.

CERTIFICATIONS, HONORS AND AWARDS

- **1st prize**, International Quant Championship 2024 – University Round (37,000+ participants) 2024
- **9th place**, International Quant Championship 2024 – National/Regional Round 2024

ACADEMIC ACTIVITIES

- VIASM Summer School on Mathematical Physics 2026, HCMUS Aug. 2026
- Summer School on Mathematical Biology 2025, VIASM Jul.–Aug. 2025
- Samsung Innovation Campus, Artificial Intelligence Course 2024

ADDITIONAL INFORMATION

Languages: Vietnamese (Native), English (TOEIC 890 – L/R 460/430)

Interests: Quantitative Research, Machine Learning, Factor Investing, Mathematical Modelling